DETECTING RELATIVE DUALIZING COMPLEXES ON AN ETALE COVERING BY AFFINES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Strong-flag prefilter hit re-read per the 2026-10-14 12:4x THIN-criterion patch (pb>=800 / pw>=100): the proof body was read in full at source; the main line is complete and the flagged phrase covers routine details, quoted verbatim in the editorial note below. Verbatim, proof line 2174: 'some details omitted' after replacing the covering by a disjoint union, and proof line 2210: 'small detail omitted' after the pullback of the vanishing statement; both cover routine bookkeeping in an otherwise complete descent argument.

Editorial note. The label lemma-covering-enough names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. Strong-flag prefilter hit re-read per the 2026-10-14 12:4x THIN-criterion patch: proof body (proof_lines 2158-2226, pb=2842, pw=383) read in full; main line complete.

Editorial note. Editorial note (verbatim): 'some details omitted' (replacing the given covering by a single disjoint-union covering) and 'small detail omitted' (pullback of the vanishing of negative self extensions).

Editorial note. Pre-open gate: python operations/agent-authoring/tools/precard_gate.py –source spaces-duality.tex –label lemma-covering-enough –start 2140 –end 2226 –local-id 28177 returned RESULT=OK. Cross-chapter same-name check: not carded in any other chapter.

Editorial note. The statement and proof are copied from the pinned chapter file; the cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Affine etale coverings refining a given covering. Topologies, the lemma producing a finite etale covering by affines refining a given etale covering. Allows replacing the given covering by a single surjective etale morphism of affines.
- Uniqueness of the relative dualizing complex and its vanishing of negative self extensions. Same chapter, the uniqueness lemma for relative dualizing complexes and the lemma on the vanishing of negative self extensions. Produces the comparison isomorphism over the covering and shows it agrees on the double pullback, so that it descends.

Date: stacks-project:spaces-duality.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Descent of maps and objects along a covering with vanishing negative self extensions. Simplicial Spaces, the lemma on gluing objects along a covering when negative self extensions vanish. Descends the comparison isomorphism to an isomorphism on the source.

Source reference. Original reference 0E5X: sources/stacks-spaces-duality/pinned-original-source.tex, source lines 1978–1978. Prerequisite pointer invoked at proof line(s) 2157; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E2G: sources/stacks-spaces-duality/pinned-original-source.tex, source lines 2054–2054. Prerequisite pointer invoked at proof line(s) 2218; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 0E2D: sources/stacks-spaces-duality/pinned-original-source.tex, source lines 2071–2071. Prerequisite pointer invoked at proof line(s) 2208; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DL9: sources/stacks-chow/tags.txt, source lines 17039–17039. Prerequisite pointer invoked at proof line(s) 2225; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0218: sources/stacks-chow/tags.txt, source lines 2661–2661. Prerequisite pointer invoked at proof line(s) 2167; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let $X \to Y$ be a proper, flat morphism of algebraic spaces which is of finite presentation. Let $(\omega^\bullet, \tau)$ be a pair consisting of a Y-perfect object of $D(\mathcal{O}_X)$ and a map $\tau : Rf_*\omega^\bullet \to \mathcal{O}_Y$. Assume we have cartesian diagrams*

$$\begin{array}{ccc} X_i & \xrightarrow[g_i']{} & X \\ {\scriptstyle f_i}\downarrow & & \downarrow{\scriptstyle f} \\ Y_i & \xrightarrow{g_i} & Y \end{array}$$

with Y_i affine such that $\{g_i : Y_i \to Y\}$ is an étale covering and isomorphisms of pairs $(\omega^\bullet|_{X_i}, \tau|_{Y_i}) \to (a_i(\mathcal{O}_{Y_i}), Tr_{f_i,\mathcal{O}_{Y_i}})$ as in Definition 0E5X. Then $(\omega^\bullet, \tau)$ is a relative dualizing complex for X over Y.

Proof. Let $g : Y' \to Y$ and X', f', g', a' be as in Definition 0E5X. Set $((\omega')^\bullet, \tau') = (L(g')^*\omega^\bullet, Lg^*\tau)$. We can find a finite étale covering $\{Y'_j \to Y'\}$ by affines which refines $\{Y_i \times_Y Y' \to Y'\}$ (Topologies, Lemma 0218). Thus for each j there is an i_j and a morphism $k_j : Y'_j \to Y_{i_j}$ over Y. Consider the fibre products

$$\begin{array}{ccc} X'_j & \xrightarrow[h_j']{} & X' \\ {\scriptstyle f'_j}\downarrow & & \downarrow{\scriptstyle f'} \\ Y'_j & \xrightarrow{h_j} & Y' \end{array}$$

Denote $k'_j : X'_j \to X_{i_j}$ the induced morphism (base change of k_j by f_{i_j}). Restricting the given isomorphisms to Y'_j via the morphism k'_j we get isomorphisms of pairs $((\omega')^\bullet|_{X'_j}, \tau'|_{Y'_j}) \to (a_j(\mathcal{O}_{Y'_j}), \mathrm{Tr}_{f'_j, \mathcal{O}_{Y'_j}})$. After replacing $f : X \to Y$ by $f' : X' \to Y'$ we reduce to the problem solved in the next paragraph.

Assume Y is affine. Problem: show $(\omega^\bullet, \tau)$ is isomorphic to $(\omega^\bullet_{X/Y}, \mathrm{Tr}) = (a(\mathcal{O}_Y), \mathrm{Tr}_{f, \mathcal{O}_Y})$. We may assume our covering $\{Y_i \to Y\}$ is given by a single surjective étale morphism $\{g : Y' \to Y\}$ of affines. Namely, we can first replace $\{g_i : Y_i \to Y\}$ by a finite subcovering, and then we can set $g = \coprod g_i : Y' = \coprod Y_i \to Y$; some details omitted. Set $X' = Y' \times_Y X$ with maps f', g' as in Definition 0E5X. Then all we're given is that we have an isomorphism

$$(\omega^\bullet|_{X'}, \tau|_{Y'}) \to (a'(\mathcal{O}_{Y'}), \mathrm{Tr}_{f', \mathcal{O}_{Y'}})$$

Since $(\omega^\bullet_{X/Y}, \mathrm{Tr})$ is a relative dualizing complex (see discussion following Definition 0E5X) there is a unique isomorphism

$$(\omega^\bullet_{X/Y}|_{X'}, \mathrm{Tr}|_{Y'}) \to (a'(\mathcal{O}_{Y'}), \mathrm{Tr}_{f', \mathcal{O}_{Y'}})$$

Uniqueness by Lemma 0E2D for example. Combining the displayed isomorphisms we find an isomorphism

$$\alpha : (\omega^\bullet|_{X'}, \tau|_{Y'}) \to (\omega^\bullet_{X/Y}|_{X'}, \mathrm{Tr}|_{Y'})$$

Set $Y'' = Y' \times_Y Y'$ and $X'' = Y'' \times_Y X$ the two pullbacks of α to X'' have to be the same by uniqueness again. Since we have vanishing negative self exts for $\omega^\bullet_{X'/Y'}$ over X' (Lemma 0E2G) and since this remains true after pulling back by any projection $Y' \times_Y \ldots \times_Y Y' \to Y'$ (small detail omitted – compare with the proof of Lemma 0E2D), we find that α descends to an isomorphism $\omega^\bullet \to \omega^\bullet_{X/Y}$ over X by Simplicial Spaces, Lemma 0DL9. □